

Arjun Vasavan

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Technical Summary

Embedded Systems Engineer skilled in C/C++, Linux system programming, and bare-metal development on PIC18F4580. Delivered 15+ projects spanning network protocols, data structures, and automotive CAN systems during advanced training at Emertxe, Bangalore.

Technical Skills

Languages: C, C++

Embedded Platforms: Linux distributions (Ubuntu)

Embedded Controllers: PIC18F4580 (GPIO, Timers, Interrupts, UART, SPI, I2C, CAN), LCD interfacing

Data Structures: Linked Lists, Arrays, Stacks, Queues, Trees, Hash Table

System Programming: Linux system calls, IPC mechanisms (Pipe, FIFO, Shared memory), Network programming (TCP/UDP sockets), pthreads (Multi-thread programming)

Developer Tools: Vim, Neovim, Makefiles, GCC, GDB (breakpoints, watch variables, source view), Git

Cloud & IoT: Arduino IoT Cloud, ThingSpeak

Experience

Emertxe Information Technologies 📄

Embedded Systems Trainee

Jun 2025 – Present

Bangalore, India

- Completed advanced embedded systems curriculum covering Linux system programming, network protocols, and bare-metal development. Delivered 15+ projects including a TCP remote process manager, TFTP protocol implementation, and automotive CAN Bus communication system.

Education

College of Engineering Chengannur

B.Tech in Electronics & Communication Engineering, Minor: Machine Learning

Graduated Jun 2025

Chengannur, Kerala

St. Thomas Higher Secondary School

Higher Secondary Education

Graduated Jun 2021

Pathanamthitta, Kerala

Contributions and Achievements

Spot Award for Outstanding Performance 📄

Emertxe Information Technologies

Oct 2025

Bangalore, India

- Received Spot Award for exceeding course requirements by completing additional C module projects beyond the core curriculum

Essential Mathematics for Machine Learning 📄

NPTEL Certification

Oct 2023

IIT Roorkee

Projects

Linux Remote Process Manager (Client–Server) |

Mar 2026

- Built a TCP-based remote process diagnostics tool to inspect memory, CPU, file descriptors, and ports of Linux processes on a remote machine – equivalent to a lightweight remote `htop`.
- **Technologies:** C, Linux System Programming, TCP Sockets, `/proc` Filesystem, Signals, `termios`, Makefiles
- **Key Challenges:** Mapped sockets to ports by resolving inodes from `/proc/<pid>/fd` against `/proc/net/tcp`; built modular command dispatch using function pointers.

Car Black Box System |

Apr 2026

- Built a PIC18F4580-based automotive black box that logs speed, gear, timestamp, and collision events to external EEPROM with circular buffer logic supporting 10 persistent log entries across power cycles.
- **Technologies:** Embedded C, PIC18F4580, I2C, ADC, UART, Timer Interrupts, LCD Interface, DS1307 RTC, External EEPROM
- **Key Challenges:** Implemented circular buffer with wrap-around and power-cycle persistence using a magic number init flag; resolved EEPROM write timing by inserting busy-wait delays to prevent stale reads; handled BCD-to-decimal conversion for accurate RTC time display.

CAN Bus Communication System (Automotive Use-Case) |

Apr 2026

- Implemented CAN protocol communication between PIC18F4580 nodes for automotive data exchange (temperature, speed) with LCD interface for real-time monitoring, Node ID configuration, and persistent storage.
- **Technologies:** Embedded C, CAN Protocol, PIC18F4580, Interrupts, LCD Interface
- **Key Challenges:** Engineered CAN arbitration and message prioritization in a broadcast system without explicit addressing; designed persistent node configuration with real-time operational behavior.

Trivial File Transfer Protocol (TFTP) |

Jan 2026

- Implemented UDP-based TFTP from scratch per RFC 1350, supporting file upload (WRQ) and download (RRQ) with opcode processing, block numbering, ACK synchronization, and termination logic.
- **Technologies:** C, UDP Sockets, Linux System Calls, Network Protocol Design, Makefiles
- **Key Challenges:** Fixed packet corruption from structure padding using `#pragma pack(1)` and `memcpy`; enforced correct network byte order via `htons/ntohs`.

Minishell |

Dec 2025

- Built a Unix-like interactive shell supporting internal and external commands with parsing, process creation, piping, and I/O redirection.
- **Technologies:** C, Process Management, System Calls, Signal Handling, Makefiles
- **Key Challenges:** Implemented process creation, signal handling, and exit status management via system calls; engineered edge case handling for multiple pipes and command chaining.

Red-Black Tree Implementation |

Mar 2026

- Implemented a self-balancing Red-Black Tree in C from scratch with insertion, deletion, search, and min/max at guaranteed $O(\log n)$ via left/right rotations and color-based balancing.
- **Technologies:** C, Data Structures, Pointers, Dynamic Memory Allocation, Makefiles
- **Key Challenges:** Resolved complex deletion using double-black and sibling-based rebalancing; enforced pointer safety and eliminated segfaults via robust NULL handling and parent pointer updates.

Source to HTML Syntax Highlighter |

Apr 2026

- Built a C source file to HTML converter that tokenizes C code into classified tokens (comments, strings, preprocessor directives, keywords, numbers) and generates syntax-highlighted HTML output.
- **Technologies:** C, Finite State Machine, Tokenization, File I/O, HTML Generation, Function Pointers, Makefiles
- **Key Challenges:** Designed modular architecture separating parser and output layers using function pointer callbacks; implemented FSM-based tokenizer handling escape sequences, nested block comments, and keyword classification without external libraries.